

只看一小角，你猜得中嗎？
Can you guess from just one corner?

老師藏起一件物件，只露出一小角
The teacher hides an object and shows just one corner

是筆？擦膠？還是……猜錯也沒關係！
A pen? An eraser? A wrong guess is welcome!

你用了甚麼線索：顏色、邊緣、形狀？
Which clues helped: colour, edges or shape?

換個背景，還認得到嗎？

A new background: still recognisable?

同一支筆，換背景、轉角度、拉遠一點

The same pen: change its background, angle or distance

你仍認得，但模型的預測會一樣嗎？

You still recognise it. Will the model give the same prediction?

先寫下你的猜測，稍後來「考考」模型

Write your guess. We will challenge the model later

今天教電腦認識三個類別

Today we teach three classes

筆、擦膠、空背景

Pen, eraser, empty background

用貼好標籤的相片，教模型分辨這三類

Teach these classes with labelled images

模型只會在我們設定的類別中作選擇

The model chooses among the classes we set

回歸猜數值，分類猜類別

Regression predicts a number; classification predicts a class

上兩堂：每天溫習時間，預測考試成績

Previously: study time predicts an exam score

今堂：輸入一張圖片，預測「筆／擦膠／空背景」

Today: an image predicts pen, eraser or empty background

一個輸出是數值，另一個輸出是類別

One output is a number; the other is a class

圖片也是數據

A picture is data too

數位圖片由很多小色點組成，稱為「像素」

A digital image contains tiny coloured pieces called pixels

你看見一支筆；電腦接收的是圖像數據

You see a pen; the computer receives image data

改角度或背景，收到的圖像數據也會改變

Change the angle or background, and the image data changes

CNN：從局部圖像學習模式

CNN: learning patterns from parts of an image

先看局部圖像，可能學到邊緣、紋理等模式

Local parts can reveal patterns such as edges and textures

把不同模式組合起來，幫助作分類

Combining patterns helps classify images

我們只需理解想法，不需計算像素或推導公式

Understand the idea. No pixel calculations or formulas needed

借用一個已學過圖像的模型

Reuse a model that has already learned image features

Teachable Machine 借用已訓練的圖像模型

Teachable Machine reuses a trained image model

再用我們的相片，建立自己的分類部分

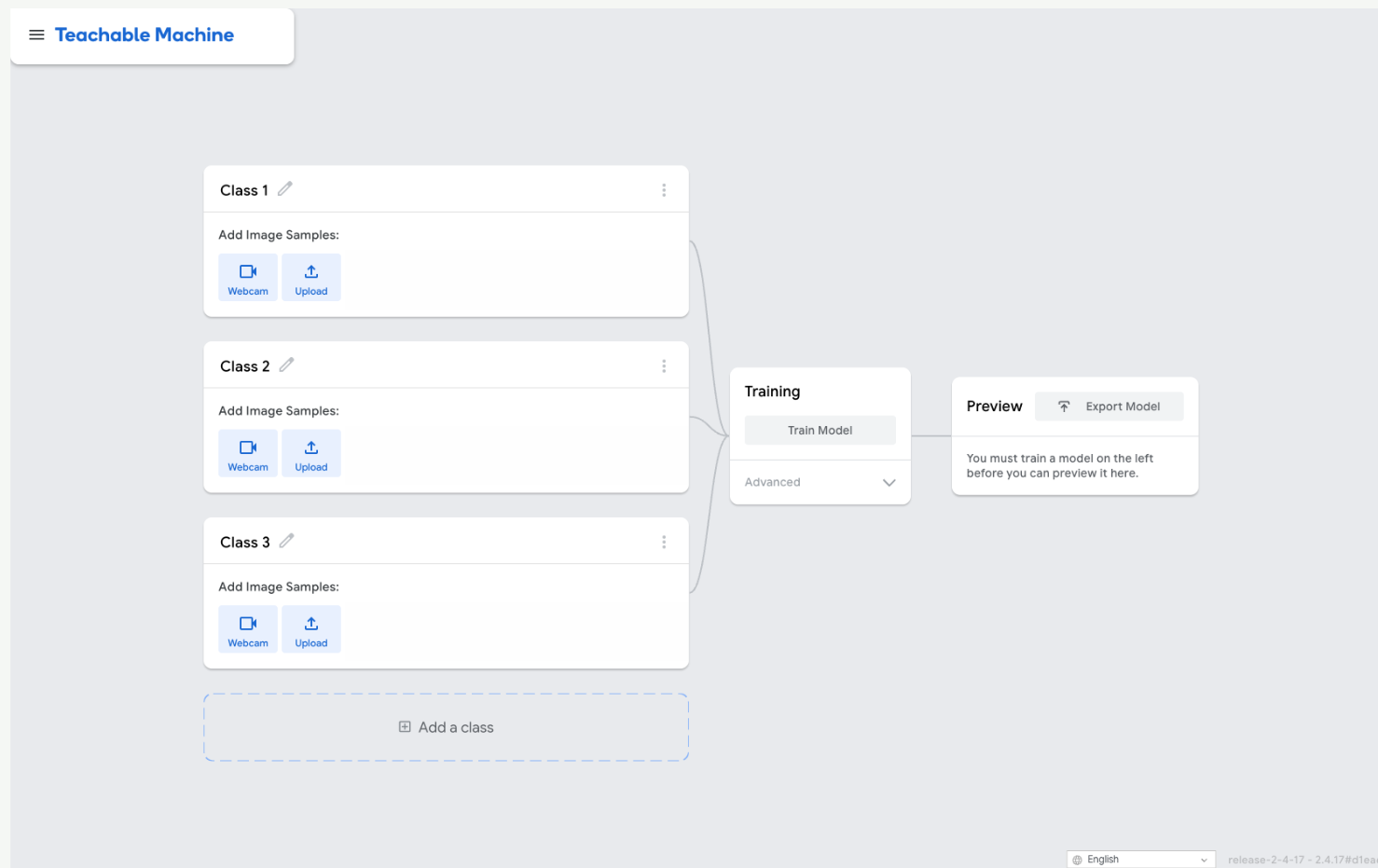
Our images teach the classifier our chosen classes

今天要花心思選好例子，不需從零編寫 CNN

Today we focus on good examples, without coding a CNN from scratch

開啟你的圖像專案

Open your image project



三個類別

Three classes

加相片

Add images

訓練，再預覽

Train, then preview

訓練相片的「食譜」

A recipe for training images

每類目標至少 20 張，數量大致相近

Aim for 20 or more per class, with similar counts

改角度、距離、背景；試不同的筆和擦膠

Vary angle, distance and background; try different pens and erasers

一直拍同一角度，可能只讓模型「背熟那一幕」

Repeated identical views may teach only that one scene

按訓練之前，先留起測試圖片
Reserve test images before training

訓練圖片：給模型學習

Training images: examples for learning

老師備妥 A：每類 4 張，用來第一次檢查

Teacher prepares A: four new images per class

B：另留每類 4 張，改善後才用；兩批難度相近

B: another four per class, used after improvement; similar difficulty

Train Model：開始學習你的類別

Train Model: learn your classes

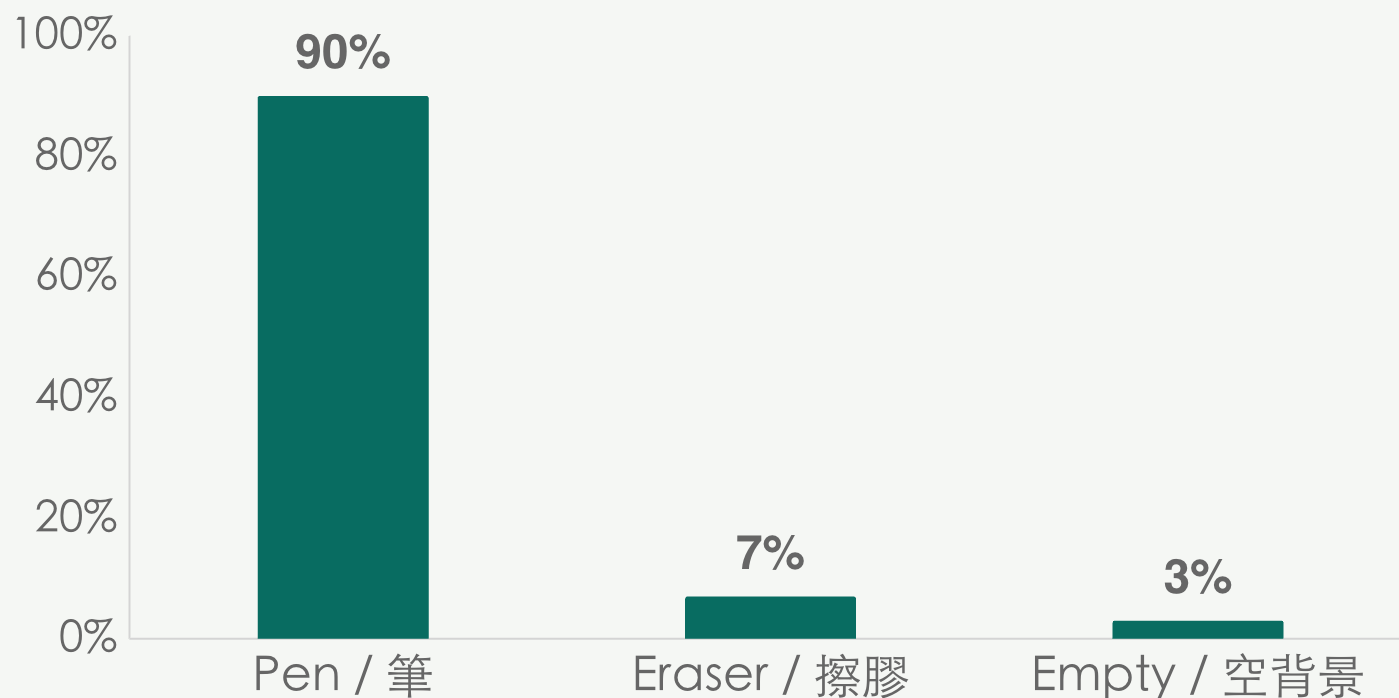
先檢查三類是否都有相片，再按 Train Model
Check that every class has images, then select Train Model

完成後，用 Preview 試 A 的新圖片
After training, use Preview with the new images in set A

記錄真實類別及預測；若 A 全對，可寫「未見錯誤」
Record actual and predicted classes. If A is all correct, write “none found”

模型「很有信心」，也可能答錯

A confident-looking model can be wrong



實際拿的是擦膠！

It is actually an eraser!

90% 不是測試準確率

90% is not test accuracy

這是示例畫面；要用新圖片的實際結果來檢查 / Illustration: check against actual results on new images

考完 A，改善，再考新的 B
Test A, improve, then try fresh B

看看 A 的錯誤：哪些角度、距離或背景較困難？
Inspect mistakes on A. Which views were difficult?

補充合適的訓練圖片，再訓練一次
Add useful training images, then train again

最後才使用 B，記每類答對多少張；不需計百分比
Use B last. Record correct counts per class; no percentages needed

突擊小考：拿一件沒教過的物件

Surprise challenge: an object never taught

換成水樽或間尺，先猜模型會選哪一類

Try a bottle or ruler. Guess which class the model will choose

它只有三個選項，不能自動新增「水樽」這一類

It has three choices; it does not automatically create a bottle class

空背景只代表教過的背景例子，不是所有未知物件

Empty background means learned background examples, not every unknown object

你的偵探報告

Your detective report

你改了哪些訓練例子？模型有甚麼改變？

What training examples did you change? What happened?

為甚麼要用模型未見過的新圖片來測試？

Why test with new images the model has not seen?

模型猜得很有信心，還需要人檢查嗎？

If the model looks confident, do we still need to check?